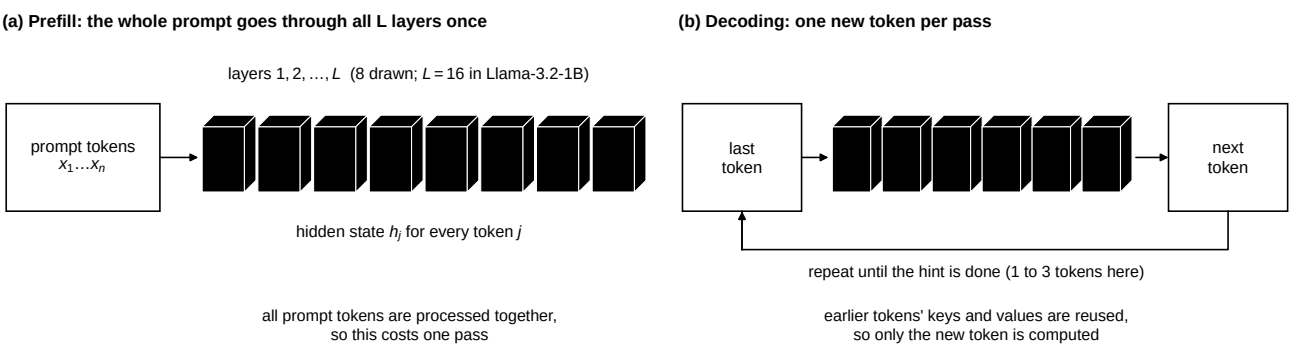
(a) Prefill: the whole prompt goes through all L layers once
layers 1, 2, …, L (8 drawn; L = 16 in Llama-3.2-1B)
prompt tokens
$x_1 \ldots x_n$
hidden state h_j for every token j
all prompt tokens are processed together,
so this costs one pass
(b) Decoding: one new token per pass
last
token
next
token
repeat until the hint is done (1 to 3 tokens here)
earlier tokens' keys and values are reused,
so only the new token is computed